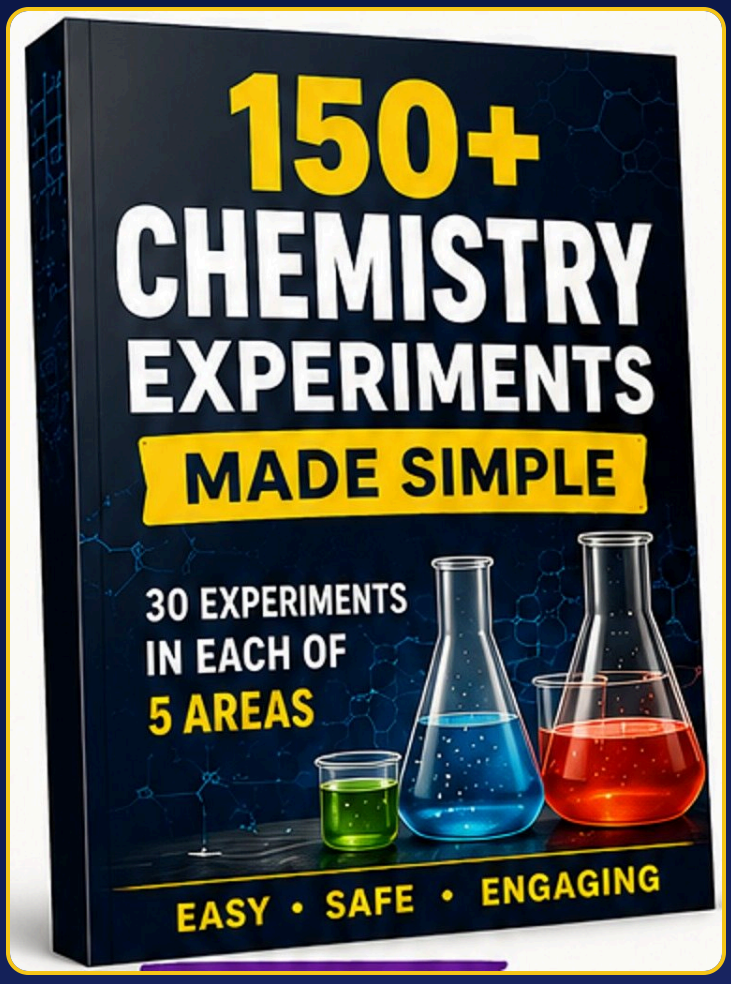
BONUS 2

Classroom Lab Safety Guide + Parent Permission Form

Protocols by experiment type, pre-lab checklist and permission form template



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Purpose and How to Use This Guide

> *Hands-on chemistry works best when every student feels safe, and every adult in the room knows exactly what to do. This guide gives you simple, repeatable safety routines for the 150+ experiment cards, plus a ready-to-adapt parent permission form.*

All 150+ experiments in this pack use household materials and low-risk methods. There are no strong acids or bases, no open flames except an adult-handled candle in a few clearly marked cards, and no reactive metals. Still, **low risk is not no risk**. Eye splashes, spills, allergies and hot water are the real-world issues in a classroom, and a calm routine prevents almost all of them.

> **Important: this guide supports, it does not replace, your policies**
>
> This guide is a teaching aid. It does not replace your school or district safety policy, your state or local regulations, or the guidance of your administration and school nurse. If anything here conflicts with your policy, follow your policy. Have your administration review this document before you use it with students.

How to use it

1. **Before the school year:** read the whole guide once, adapt the permission form, and send it home with the first-week packet.
2. **Before each experiment:** read the card's safety note, find the matching protocol below, and run the teacher pre-lab checklist (about 3 minutes).
3. **At the start of the first lab:** teach the 10 Golden Rules and have students take the pre-lab safety quiz. Keep the signed quiz on file.
4. **During labs:** post the Golden Rules where everyone can see them and use the same clean-up routine every time.
5. **If something happens:** use the spills and first aid table, then fill in the incident report form.

Every experiment card has a safety line and an adult-handling flag. The protocols in this guide are the longer version of those lines. When a card says *adult handling*, a teacher or another adult does that step and students observe from a safe distance.

The 10 Golden Rules of the Classroom Lab

Print this table as a poster, one rule per row, and keep it visible at every station. Teach one or two rules per week at first, then review all ten before every lab.

#	RULE	WHY IT MATTERS
1	Goggles on before materials come out, and stay on until everything is cleaned up.	Eyes are the most vulnerable part of the body. Most lab injuries are splashes.
2	Listen to the whole set of directions before you touch anything.	Most mistakes happen when students start early.
3	Never taste, eat or drink anything in the lab area, and never put hands near your face.	Lab items are not food, even when the ingredients look like food.

#	RULE	WHY IT MATTERS
4	Follow the card exactly. Do not change amounts or mix extra things.	Unplanned mixtures are the main cause of surprises.
5	Adults handle hot water and flames. Students never do.	Burns are serious, and they are easy to prevent.
6	Never seal a container that makes gas, and never point an opening at anyone.	Gas needs somewhere to go. Open containers only.
7	Walk, do not run. Keep bags and coats away from stations.	A clear floor prevents spills and falls.
8	Tell the teacher right away about every spill, break, cut, splash or odd smell.	Small problems are easy to fix. Hidden ones are not.
9	Smell by wafting only: wave the air toward your nose with your hand.	Direct sniffing can irritate your nose and throat.
10	Wash hands with soap and water after every lab, and leave your station cleaner than you found it.	It protects you, your classmates and the next class.

Personal Protective Equipment (PPE) and How to Teach It

PPE is the last line of defense, not the first. Good procedures come first, and PPE protects when something goes wrong. Teach each item as a habit, not as a punishment.

ITEM	WHAT TO USE	HOW TO TEACH IT
Splash goggles	Indirect-vent splash goggles that seal around the eyes (not regular glasses). Wipe them with soap and water between classes.	Demonstrate a correct fit: snug on the face, strap tight, no gaps. Students who wear prescription glasses wear goggles over them, or use goggles made for that purpose.
Apron or old shirt	A lab apron, or an old long-sleeved button shirt worn backwards.	Explain that it protects clothes and skin from splashes and stains. Roll sleeves up or keep them snug.
Closed shoes	Closed-toe shoes that cover the whole foot.	Sandals and open shoes are not allowed on lab days. Send a reminder home the day before.
Hair tied back	Long hair tied back, loose jewelry and dangling items removed or tucked in.	Make a quick check at the door. Keep a few spare hair ties on hand.
Gloves (when needed)	Disposable gloves, preferably nitrile, for cleaning up, handling irritants or when a student has sensitive skin.	Ask about latex sensitivity first. Gloves are not a substitute for hand washing. Remove them inside out and wash hands afterward.

> **Teaching tip: make PPE routine**
>
> Create a goggle station at the door. Students pick up goggles when they enter, put them on at their seats, and turn them in at the end of class. A 20-second ritual repeated every lab becomes automatic within two weeks.

Room Setup and Class Management

Stations

- Set up one tray or bin per group with everything students need. A tray catches spills and keeps the work area small.
- Cover tables with a plastic tablecloth or paper. Keep a roll of paper towels and a spill sponge at every station.
- Place a labeled waste cup or bucket at each station for liquids and a bag for solids.
- Keep the teacher's materials table separate from the student area. Anything with an irritant or a heat source stays there.

Group roles

ROLE	JOB
Materials Manager	Picks up and returns the tray. Is the only person who walks to the supply table.
Measurer	Measures and pours, always slowly and at the table.
Recorder	Writes observations, times and results on the sheet.
Safety Officer	Checks goggles, reads the safety note aloud, and calls the teacher if anything is off.

Rotate roles every lab so everyone practices each job. Groups of four work well. In groups of three, combine Recorder and Safety Officer.

Traffic flow

- Mark one path to the supply table and another to the sink. Students walk one at a time.
- Keep aisles clear. Bags and coats go in a corner or on hooks, never under tables.
- Use a quiet signal (a raised hand or a chime) to pause the whole room instantly.

The clean-up routine

1. Signal: **freeze and hands up** when you announce clean-up.
2. Pour liquids into the waste cup as directed by the disposal table.
3. Wipe the tray and table with a damp paper towel or sponge.
4. Return materials to the tray, and the tray to the supply table.
5. Remove goggles last, then wash hands for 20 seconds.
6. Safety Officer signals the teacher for a final look. Stations are cleared one at a time.

Leave 5 to 8 minutes for clean-up in every plan. A rushed clean-up is when most spills and mix-ups happen.

Protocols by Experiment Type

Find the type of experiment on your card and follow the matching protocol. Many experiments fit more than one type, so combine the protocols when needed.

Acid/Base and Indicator Experiments

These experiments use vinegar, lemon juice, baking soda, washing soda solutions and natural indicators such as red cabbage juice. All are mild, but they can still sting eyes and irritate sensitive skin.

- Splash goggles on for every step, including when only observing.
- Use small quantities. Measure into cups, never pour directly from a large bottle.
- Never mix household cleaners to "make something stronger." Only the materials on the card are used.
- Never taste a test solution. Use clean droppers and do not share them between solutions.
- Rinse skin with plenty of water if any liquid touches it.

Reactions That Produce Gas or Foam

Baking soda and vinegar, yeast and peroxide, and similar reactions release carbon dioxide or oxygen. Gas is harmless when it can escape, but it can burst a closed container.

> **Open containers only, never sealed**
>
> Never cap, cork, tape or seal a container where gas is being produced. Balloons on bottle necks are acceptable only where the card states it, using small amounts and a plastic bottle that is not damaged. Point openings away from every face, including your own.

- Do the reaction on a tray, inside a larger container, so foam stays contained.
- Use the exact amounts on the card. More material does not mean a better show, only a bigger mess.
- Keep faces away from the opening. Observe from the side at arm's length.
- Do not carry a reacting container around the room. Keep it still on the tray.

Heat and Hot Water

Warm water speeds up many reactions and dissolves materials faster. It is also the most common source of burns. Treat any water above warm-tap temperature as an **adult-handled** material.

- Only an adult heats water, pours hot water and moves containers that hold it.
- Use a thermometer and keep the water below 140 degrees F (60 degrees C) wherever possible. Water at this temperature or lower is less likely to cause a serious scald.
- Use sturdy heat-safe containers. Never use thin glass, cracked containers or containers with cracks or chips.
- Students stay seated while the adult pours. Move away tables and bags first.
- Let hot items cool completely before students touch them.

Candles and Flames (Adult Only, If Used)

A small number of cards use a single tea light or birthday candle. If your school does not allow flames, skip these cards or use a battery-powered tea light as a substitute for the demonstration. Never invent a flame experiment that is not on a card.

- The adult lights, holds and extinguishes the candle. No student touches matches or lighters.
- Use a stable, non-flammable base, such as a metal tray or a ceramic plate, on a cleared table.
- Tie back hair, tuck in sleeves and remove loose papers from the area.
- Keep a damp towel and a bucket of water or a fire extinguisher within reach, and know where the nearest one is.
- Keep students at least an arm's length away. Never leave a flame unattended.
- Follow your school's policy on open flames. If in doubt, ask your administration first.

Cutting and Sharp Tools

- An adult does all cutting with knives, craft blades or scissors that need force.
- Students may use blunt-tip scissors only for paper and tape.
- Cut away from the body on a stable cutting board, never in the hand.
- Dispose of any broken glass or sharp items in a sturdy container, never loose in the trash. Do not use glass containers that can crack.
- Keep sharp tools on the teacher's table and count them before and after class.

Food-Based Experiments

Some cards use food items such as eggs, milk, flour, strawberries, cabbage or yeast. These are still lab items.

> **Not eaten, ever**
>
> Nothing prepared in a science lesson is eaten in the classroom. The only exception is a card that explicitly describes a kitchen-style activity made with clean food-grade tools away from the lab area, and even then follow your school's food policy. When in doubt, do not eat it.

- Use separate, clearly labeled food-grade items for any food-based experiment. Never reuse items that touched lab materials for food.
- Run an allergy check before the lab (see the allergy screening section). The most common allergens in these cards are eggs, milk, wheat, nuts and strawberries.
- Wash hands before and after handling food items, and clean surfaces with soap and water.
- Throw away food items that have been used. Do not take them home.
- Never use a kitchen appliance in the classroom unless your school allows it and an adult operates it.

Crystal Growing and Long-Term Experiments

- Label every jar with group name, date and experiment title using masking tape and a marker.
- Keep jars on a tray, in a place students cannot knock them over, away from food and drinks.

- Warm saturated solutions are adult-handled. Let them cool fully before students touch them.
- Keep jars covered loosely with a paper towel (not sealed) so dust stays out and vapor can escape.
- Check weekly for mold, leaks or cracked containers. Dispose of anything doubtful as directed in the disposal table.
- Crystals are small, hard and sharp-edged. Do not allow students to put them in their mouths or take them home without permission.

Chemicals That Are Skin or Eye Irritants

Some common household items are safe in normal use but can irritate eyes or skin in a classroom setting. Treat these with care.

MATERIAL	MAIN CONCERN	HOW TO HANDLE IT
Rubbing alcohol	Irritates eyes, dries skin, and is flammable.	Use small quantities, keep it far from any flame or heat source, and keep the room ventilated. Adults dispense it.
Hydrogen peroxide (3% only)	Irritates eyes and skin, and bleaches clothes.	Use only the 3% pharmacy strength. Store it in its original dark bottle. Never use stronger solutions. Rinse skin right away.
Vinegar	Stings eyes and cuts, and can irritate sensitive skin.	Goggles on. Rinse with water if it touches skin.
Epsom salt	Can irritate eyes, and is not safe to swallow in quantity.	Keep it away from mouths. Wash hands after use.
Iodine tincture	Stains skin and clothes, irritates eyes, and is harmful if swallowed.	Adult dispenses it a few drops at a time. Check for allergies and sensitivity first. Gloves recommended. Never use it with students who have iodine sensitivity.

Electricity (Batteries Only)

- Use only small batteries, such as AA, AAA or a single 9 V battery, and only for experiments on the card.
- Never use wall outlets, extension cords, power strips, chargers or any mains-powered equipment for an experiment.
- Never connect the two ends of a battery directly with a wire. It heats quickly and can burn.
- Do not use salt water with a battery for any reason other than what the card describes.
- Remove batteries and tape them before storage, and dispose of old batteries at a collection point, not in the trash.

Biology and Microbe-Adjacent Experiments

Yeast, bread and fruit decay, and mold observation are low risk when done correctly. The main risks are allergies and breathing in spores.

- Use dry yeast and warm (never hot) water. Yeast is not a hazard, but wash hands afterward.
- **Do not open sealed mold containers.** If mold grows in a sealed bag or jar, leave it sealed, observe through the walls, and discard the whole container sealed.

- Keep mold experiments away from students with asthma or mold allergies. Check with parents before the activity.
- Never taste or smell mold. Keep these projects out of kitchens and break rooms.
- Wash hands after handling any biological material and disinfect the surface with soap and water.

Allergy and Sensitivity Screening

Allergies are the biggest hidden risk in a materials-based lab. Screen early, record privately and share only what is needed with the adults who need it.

1. Send the parent permission form (below) at the start of the year. It includes a checklist for common sensitivities.
2. Review the forms before each lab and check the card's materials list for matches.
3. Check with the school nurse for any medical plans that apply. Keep a private list of affected students and what they need.
4. Offer a safe alternative when a student is sensitive: a different material, an observer role, or a partner who handles the item.
5. Keep allergy information confidential. Do not announce a student's condition to the class.

COMMON CONCERN	WHERE IT MIGHT APPEAR	ALTERNATIVE
Eggs	Eggshell and egg experiments	Use a plastic or wax model, or have the student observe.
Milk	Milk and protein experiments	Use a plant-based milk that is allowed under the student's plan.
Wheat (gluten)	Flour, bread and dough	Use cornstarch where the card allows.
Nuts	Rare, such as cooking oils and spreads	Replace the item or remove the card from that class.
Strawberries and other fruit	Pigments and indicators	Use a different fruit or vegetable, such as red cabbage.
Latex	Gloves and balloons	Use nitrile gloves and a non-latex alternative.
Asthma or strong smells	Vinegar, alcohol, mold	Ventilate, seat the student by a window and provide an observer role.

Spills and First Aid

> **Always**
>
> Follow your school's emergency procedures and call your local emergency number when needed. This table is quick reference for the teacher. It does not replace school nurse guidance or your school's medical policy.

SITUATION	FIRST RESPONSE	THEN
Eye splash	Go to the eyewash or a clean sink immediately. Hold the eyelids open and flush with a gentle stream of clean water for at least 15 minutes. Remove contact lenses if easily possible.	Notify the school nurse and the administration. Follow your school's emergency procedures and call your local emergency number when needed.
Skin contact	Remove any contaminated clothing or jewelry. Rinse the skin under running water for at least 15 minutes.	Notify the nurse. File an incident report. Inform the parent or guardian.
Ingestion (swallowed material)	Do not make the student vomit. Rinse the mouth with water and keep the student calm and seated. Have the product container on hand.	Follow your school's emergency procedures and call your local emergency number when needed. Notify the nurse and the parent or guardian.
Burns (heat)	Cool the burned area under cool (not ice-cold) running water for at least 10 minutes. Do not apply ointments, butter or ice.	Notify the nurse. Follow your school's emergency procedures and call your local emergency number when needed for blisters, large areas or burns on the face or hands.
Cuts	Wash your hands, apply gentle pressure with a clean cloth or sterile gauze, and raise the area if possible.	Notify the nurse. Follow your school's emergency procedures and call your local emergency number when needed for deep cuts or heavy bleeding.
Liquid spill	Alert the class, keep students back, and cover the spill with paper towels. Wear gloves if the material is an irritant.	Wipe from the edges inward, bag the towels, and wash the area with soap and water. Note the spill in the log.
Broken container	Stop everyone and keep students away. An adult sweeps up the pieces with a brush and dustpan. Never use bare hands.	Place the pieces in a sturdy labeled box or container. Report the break.

Waste and Disposal

Almost everything in this pack is household-level waste. Still, a consistent routine teaches students to think about environmental impact. Always follow your school's and local waste rules first.

WHERE IT GOES	WHAT BELONGS THERE	NOTES
Down the drain with plenty of running water	Diluted vinegar solutions, baking soda solutions, dish soap solutions, salt and sugar solutions, small amounts of indicator liquid, food coloring in small amounts	Flush with water. Check your local rules before pouring any colored liquid or anything that might stain.
In the trash	Used paper towels, filter paper, plastic cups, foam, cotton swabs, dried powders, used gloves, cut or broken pieces (wrapped), batteries are the exception	Wrap sharp items securely. Seal bags before disposal.
In compost or the organic bin	Plain fruit and vegetable scraps, eggshells and used tea or coffee grounds that have not touched other chemicals	Only if your school has a compost program. Do not compost anything used with iodine, alcohol or peroxide.

WHERE IT GOES	WHAT BELONGS THERE	NOTES
Special collection (ask your administration)	Spent batteries, leftover rubbing alcohol, iodine tincture, expired peroxide, and any unknown liquids	Never pour unknown liquids down the drain. Label and store them until your school's procedure is clear.

Never mix leftover liquids from different groups into one bucket. Mixed waste can react in surprising ways. Pour each group's waste into the labeled cup, then dispose of it separately.

Chemical Storage and Labeling

- Store all materials in a locked cabinet or closet when you are not in the room. Only adults hold the key.
- Keep everything in its original container whenever possible. Never use food or drink containers for lab materials, and never store food items in the lab cabinet.
- Label every container with its contents, the date it was opened and your name. Use a permanent marker or a printed label.
- Store irritants (rubbing alcohol, peroxide, iodine) on a low shelf in a tray, away from heat and sunlight, and separate from food items.
- Keep peroxide in its original dark bottle. Keep alcohol away from any heat source.
- Do not store different cleaning products together. Keep every product in its own labeled space.
- Make an inventory sheet and check it at the start of each month. Remove anything that is expired, unlabeled or leaking.
- Keep a copy of the product label or safety information for each item in a folder in the classroom.

Pre-Lab Checklist and Student Quiz

TEACHER PRE-LAB CHECKLIST (3 MINUTES BEFORE EACH LAB)

☐ I read the experiment card and the safety note, and I know which protocol above applies.

☐ I checked the materials list against my allergy and sensitivity notes.

☐ Every student has goggles and an apron or old shirt, closed shoes, and hair tied back.

☐ Each station has a tray, paper towels, a waste cup and a spill sponge.

☐ Hot water, candles and cutting are planned for an adult only.

☐ Ventilation: windows or the door is open, or the air system is on.

☐ The sink, eyewash or water source is working and the path is clear.

☐ The first aid kit is stocked and easy to find, and I know my school's emergency procedures.

☐ Student groups and roles are assigned.

☐ The clean-up routine and a time for it are planned.

☐ I reviewed the 10 Golden Rules with the class today.

Student Pre-Lab Safety Quiz

Give this short quiz before the first lab and again at midyear. Students answer in writing, and you review the answers together.

1. What must you put on before any materials come out?
2. A container is making bubbles. Is it safe to put a lid on tightly? Why or why not?
3. Who is allowed to pour hot water?
4. How do you smell a liquid safely?
5. A friend spills a little vinegar on the table. What do you do first?
6. You get a splash in your eye. What do you do and for how long?
7. True or false: it is fine to taste a science experiment if the ingredients are from the kitchen.
8. Name two things you should do during clean-up.

#	ANSWER KEY
1	Splash goggles (and apron or old shirt, with hair tied back and closed shoes).
2	No. Gas needs a way to escape, and a closed container can burst. Use open containers only.
3	An adult (the teacher). Students never pour hot water.
4	Waft: wave the air toward your nose with your hand. Never sniff directly.
5	Tell the teacher, keep others away and cover the spill with paper towels. Do not wipe it with bare hands if it is an irritant.
6	Tell the teacher right away, go to the eyewash or sink, and flush with gentle clean water for at least 15 minutes.
7	False. Nothing from the lab is ever tasted.
8	Any two: pour liquids into the waste cup, wipe the tray and table, return materials, wash hands, remove goggles last.

Incident Report Form Template

Fill in a form for every injury, spill, breakage or near miss, even small ones. It helps you spot patterns and gives your administration the information it needs. Follow your school's own reporting process as well.

FIELD	DETAILS (FILL IN)
Date and time	
Class and period	
Teacher or adult in charge	
Student name(s) involved	
Experiment title and card ID	
What happened (facts only, in order)	

FIELD	DETAILS (FILL IN)
Where in the room it happened	
Materials involved	
First response taken (who did what)	
Nurse or administration notified (time and name)	
Emergency number called? (yes/no, time)	
Parent or guardian notified (time and by whom)	
Possible cause	
What will change next time	
Teacher signature and date	
Administrator review signature and date	

Parent Permission Form Template

> **Note to the teacher**
> This is a template. Adapt it to your school's policy and have it reviewed by your administration before sending it home. Add your school's name, local requirements and any additional language that your district needs.

Hands-On Science Activities: Parent/Guardian Permission and Information Form

School: _____ School year: _____

Student name: _____ Class / grade: _____

Teacher: _____

1. Description of activities. This year, students will take part in hands-on science activities in class. Activities include mixing safe household ingredients, observing color and temperature changes, testing acids and bases with natural indicators, growing crystals and making simple models. Activities are led by the teacher. Students wear splash goggles, follow written safety rules and are supervised at all times. An adult handles any hot water, flame or cutting. Nothing made in these activities is eaten or taken home unless the teacher says so in writing.

2. Typical household materials used. White vinegar, baking soda, table salt, sugar, dish soap, food coloring, cornstarch, flour, dry yeast, 3% hydrogen peroxide, rubbing alcohol, Epsom salt, lemon juice, red cabbage, eggs, milk, tea, coffee, paper, aluminum foil, plastic cups, balloons, small batteries, wire and ice.

3. Allergy and sensitivity disclosure. Please check anything that applies to your child:

ALLERGY AND SENSITIVITY CHECKLIST

☐ Eggs

☐ Milk or dairy

☐ Wheat or gluten

☐ Tree nuts or peanuts

☐ Strawberries or other fruit

☐ Latex (gloves, balloons)

☐ Iodine

☐ Strong smells, vinegar or alcohol

☐ Asthma or breathing sensitivity

☐ Sensitive or eczema-prone skin

☐ Mold allergy

☐ Other: _____

Details and any treatment plan: _____

4. Emergency contact information (to be completed by the school or the family, as your school requires):

FIELD	INFORMATION
Parent or guardian name	
Relationship to student	
Primary contact number	
Second emergency contact name	
Second contact number	
Doctor or medical information (optional)	
Medical conditions the school should know about	

5. Permission. I have read the description of activities above. I give permission for my child to take part in hands-on science activities. I understand that safety rules are taught and enforced, and that my child may be asked to step out of an activity if the rules are not followed. I understand that I may contact the school with questions or to change my choices at any time.

Parent/guardian name (print): _____

Parent/guardian signature: _____ Date: _____

Student signature (I will follow the safety rules): _____ Date: _____

> **Teacher note**
> Adapt to your school's policy and have it reviewed by your administration. Keep signed forms on file and check them before every lab. This form does not replace any consent or medical form that your school or district already requires.

Short Letter to Parents

Dear families,

This year, our class will explore chemistry through simple hands-on experiments. Students will see how things change, mix and react using everyday materials such as vinegar, baking soda, salt and food coloring. These activities build curiosity, careful observation and the habit of asking why.

Safety is our first priority. Every student wears splash goggles, follows a short set of safety rules and works in a small team with clear roles. A teacher handles any hot water, flame or cutting. Students do not taste or eat any materials, and everything is cleaned up with a routine we practice together.

Please complete and return the attached permission form, including the allergy and sensitivity checklist. This helps us choose safe materials for every child. If your child has a medical condition or a concern about any activity, please write it on the form and we will make a plan together.

Thank you for supporting your child's learning. Ask them about the experiment of the week, and what they saw, and why they think it happened.

Sincerely,

_____ (Teacher name) Date: _____

Next Step

You now have a complete safety system: the 10 Golden Rules, PPE routines, room setup, protocols for every kind of experiment, first aid and disposal tables, a pre-lab checklist, a quiz, an incident report and a parent permission form.

1. Adapt the permission form to your school and send it home this week.
2. Share this guide with your administration for review, and keep a copy of your own school's policy beside it.
3. Print the 10 Golden Rules and teach the first two rules at your next class.
4. Choose your first experiment card from the pack and run the pre-lab checklist before you start.

Remember: this guide supports your school's and district's safety policy. It does not replace it. When your policy and this guide differ, your policy wins.